

CSS Fonts

Complete Guide with Code Examples & Practical Usage

Introduction

CSS fonts control the typeface, size, weight, and style of text on web pages. With access to web fonts and system fonts, designers can create distinctive typography that enhances brand identity and improves readability. This comprehensive guide covers font families, web fonts, font loading, and best practices for professional text styling.

Font Properties Reference

Property	Description	Example Values
font-family	Font typeface(s)	Georgia, serif, Arial, sans-serif
font-size	Text size	16px, 1.5em, 2rem
font-weight	Text thickness	400, 700, bold, normal
font-style	Font angle	normal, italic, oblique
font-variant	Small caps effect	normal, small-caps
font-stretch	Font width	normal, condensed, expanded
font (shorthand)	All font properties	bold 18px/1.6 Arial, sans-serif

Example 1: System Font Families

Built-in fonts available on most systems.

CSS Code:

```
/* Serif fonts - formal, traditional */ p { font-family: Georgia, 'Times New Roman', serif; } /* Sans-serif fonts - modern, clean */ p { font-family: Arial, Helvetica, sans-serif; } /* Monospace fonts - code, technical */ p { font-family: 'Courier New', Courier, monospace; } /* Font stack - fallback chain */ body { font-family: Verdana, Geneva, sans-serif; }
```

HTML Usage:

```
<p style="font-family: Georgia, serif;"> Serif font example </p>
```

Example 2: Font Stacks & Fallbacks

Create fallback chains for cross-browser compatibility.

CSS Code:

```
/* Font stack - browser tries each until available */ body { font-family: 'Segoe UI', Roboto, Oxygen-Sans, Ubuntu, Cantarell, 'Helvetica Neue', sans-serif; } /* Generic fallback at end */ p { font-family: 'Custom Font', Arial, sans-serif; } /* Safe web fonts */ body { font-family: Georgia, 'Times New Roman', serif; }
```

HTML Usage:

```
<p style="font-family: 'Segoe UI', Roboto, Arial, sans-serif;"> Text with font stack </p>
```

Live Demo:

This text uses a font stack with multiple fallback options

Example 3: Google Fonts Integration

Access thousands of free web fonts.

HTML Header (Import):

```
<link rel="preconnect" href="https://fonts.googleapis.com"> <link rel="preconnect" href="https://fonts.gstatic.com" crossorigin> <link href="https://fonts.googleapis.com/css2?family=Roboto:wght@400;700&family=Playfair+Display:wght@700&display=swap" rel="stylesheet">
```

CSS Usage:

```
/* Use imported fonts */ h1 { font-family: 'Playfair Display', serif; } p { font-family: 'Roboto', sans-serif; } /* With weights */ body { font-family: 'Roboto', sans-serif; font-weight: 400; } strong { font-family: 'Roboto', sans-serif; font-weight: 700; }
```

Live Demo:

Roboto font - clean and modern

Lato font - friendly and open

Playfair Display - elegant serif

Example 4: @font-face for Custom Fonts

Load custom font files from your server.

CSS Code:

```
/* Define custom font */ @font-face { font-family: 'CustomFont'; src: url('font.woff2') format('woff2'), url('font.woff') format('woff'); font-weight: normal; font-style: normal; } /* Multiple weights */ @font-face { font-family: 'CustomFont'; src: url('font-bold.woff2') format('woff2'); font-weight: 700; } /* Use custom font */ body { font-family: 'CustomFont', Arial, sans-serif; }
```

HTML Usage:

```
<p style="font-family: 'CustomFont', Arial;"> Using custom font </p>
```

Live Demo:

Custom font loaded with @font-face

Example 5: Font Shorthand Property

Combine multiple font properties in one declaration.

CSS Code:

```
/* Shorthand: style weight size/line-height family */ p { font: 700 18px/1.6 'Arial', sans-serif; } /* Expanded equivalent */ p { font-weight: 700; font-size: 18px; line-height: 1.6; font-family: 'Arial', sans-serif; } /* With font-style */ h1 { font: italic bold 32px Georgia, serif; }
```

HTML Usage:

```
<p style="font: 700 18px/1.6 Arial, sans-serif;"> Shorthand font property </p>
```

Live Demo:

Text using font shorthand property

Example 6: Responsive Font Sizes

Scale fonts for different screen sizes.

CSS Code:

```
/* Desktop */ html { font-size: 16px; } h1 { font-size: 2.5rem; } p { font-size: 1rem; } /* Tablet */ @media (max-width: 768px) { html { font-size: 15px; } h1 { font-size: 2rem; } } /* Mobile */ @media (max-width: 480px) { html { font-size: 14px; } h1 { font-size: 1.5rem; } }
```

HTML Usage:

```
<h1 style="font-size: 2.5rem;">Title</h1>
```

Live Demo:

Small text (12px)

Normal text (16px)

Large text (24px)

Example 7: Font Variants

Special font effects like small caps.

CSS Code:

```
/* Small caps */ .small-caps { font-variant: small-caps; } /* Numeric variants */ .tabular { font-variant-numeric: tabular-nums; } /* Ligatures */ .ligatures { font-variant-ligatures: common-ligatures; }
```

HTML Usage:

```
<p style="font-variant: small-caps;"> SMALL CAPS TEXT </p>
```

Live Demo:

THIS TEXT USES SMALL CAPS VARIANT

Example 8: Font Smoothing

Control anti-aliasing for font rendering.

CSS Code:

```
/* Smooth font rendering */ body { -webkit-font-smoothing: antialiased; -moz-osx-font-smoothing: grayscale; } /* Keep subpixel rendering */ .preserve-subpixel { -webkit-font-smoothing: subpixel-antialiased; }
```

HTML Usage:

```
<body style="-webkit-font-smoothing: antialiased;"> Smooth text rendering </body>
```

Note:

Font smoothing affects rendering quality. Use antialiased for cleaner appearance, especially with light text on dark backgrounds.

Best Practices

1. Load Fonts Efficiently

```
/* Use font-display for better performance */ @font-face { font-family: 'CustomFont'; src: url('font.woff2') format('woff2'); font-display: swap; } /* Show fallback while loading */ }
```

2. Create Proper Font Stacks

```
/* End with generic fallback */ body { font-family: 'Segoe UI', Roboto, Arial, sans-serif; } /* Generic fallback */ }
```

3. Use Rem Units for Scalability

```
html { font-size: 16px; } h1 { font-size: 2.5rem; } /* 40px */ p { font-size: 1rem; } /* 16px */
```

4. Limit Font Weights

```
/* Load only needed weights */ @import url('https://fonts.googleapis.com/css2?family=Roboto:wght@400;700&display=swap'); /* Use available weights */ body { font-weight: 400; } strong { font-weight: 700; }
```

5. System Fonts for Performance

```
/* Fast loading system fonts */ body { font-family: -apple-system, BlinkMacSystemFont, 'Segoe UI', Roboto, sans-serif; }
```

Tip: Modern browsers support variable fonts, which combine multiple weight/width variations into a single file. This reduces file size while providing flexibility.

Common Issues & Solutions

Issue: Custom Font Not Loading

Problem:

```
@font-face { font-family: 'CustomFont'; src: url('font.ttf'); } /* TTF may not work on all browsers */
```

Solution:

```
@font-face { font-family: 'CustomFont'; src: url('font.woff2') format('woff2'), url('font.woff') format('woff'), url('font.ttf') format('truetype'); }
```

Issue: Font Loads Too Slowly

Problem:

```
/* Block rendering until font loads */ @font-face { font-family: 'CustomFont'; src: url('font.woff2'); font-display: block; } /* Invisible until ready */ }
```

Solution:

```
/* Show fallback immediately */ @font-face { font-family: 'CustomFont'; src: url('font.woff2'); font-display: swap; } /* Use fallback while loading */ }
```

Issue: Inconsistent Font Rendering

Problem:

```
p { font-family: 'CustomFont'; } /* Looks different on different browsers */
```

Solution:

```
p { font-family: 'CustomFont', Arial, sans-serif; -webkit-font-smoothing: antialiased; }
```

Summary

- Use **font-family** with fallback stacks
- Combine fonts into stacks ending with generic fallback
- Use **Google Fonts** for free web fonts
- Use **@font-face** for custom fonts
- Apply **font-display: swap** for performance
- Use **rem units** for scalable sizing
- Load only necessary font weights
- Test fonts across browsers and devices
- Use system fonts for fastest loading
- Optimize web fonts with modern formats (WOFF2)
- Apply font-smoothing for consistent rendering